

PERSONAL STATEMENT

There is a very famous adage, "If you think education is expensive, then try ignorance". I could not agree more. I believe research is necessary to acquire data and formulate theories, but it is just as important to know how to apply those theories and use that data in the real world. To be competent and competitive I will require a master's degree. A master's degree will give me the up-to-date tools and knowledge that are the need of the hour.

While in my third year of undergraduate study, I learnt the concepts of signal processing and microprocessors. These topics have fascinated me ever since. When I studied the fundamentals of analog and digital signals, I almost decided that Digital Signal Processing was the career for me. What got me more enthused in this area was when I learnt about microprocessors. I had the opportunity to study some of the processors during my third and fourth (final) year. That's when I realized how the two fields could be merged together by having dedicated processors. I am certain that my interest in this will increase by taking Digital Signal Processing and Microcomputer System Design as electives in my final semester of undergraduate study.

In DSP we shall learn in depth about FIR, IIR and Quantization errors. Here we shall also cover Z-transform analysis, filter design (IIR filter design and FIR filter design), DTF, FFT and its applications. As an introduction we shall learn Texas Instruments and Analog Devices special DSP Processors. In Microcomputer System Design, the Pentium Processor and PCI bus will be covered. Conceptual view of Microsoft Windows NT, Windows NT models- client server is also a part of the curriculum.

Digital signal processing is one of the brightest spots in the semiconductor business today, and one of the few deserving the title "breakthrough". Like earlier advances in microprocessors and computer memories, digital signal processing is a foundation technology with the power to transform broad areas of the electronics industry. Its impact is being felt in applications as diverse as stereo systems, cars, personal computers and cellular phones. In the next few years, digital signal processing should give rise to hundreds of new products and change what people expect from technology.

The fact that the world market in programmable Digital Signal Processor chips hit more than \$3 million in 1997, having tripled in just three years shows that Digital Signal Processor have come a long way and are here to stay. It is projected to increase to around \$14 million by the end of 2002. The future in this field is ever increasing and provides opportunity for the to-be engineers in this field.

Also being a member of the internationally acclaimed Institute of Electrical and Electronic Engineers (IEEE - currently Communication Society, next year I will be enrolled for Signal Processing Society) has given me the opportunity to keep abreast with the latest in today's advancing technology. This has also increased my interest in the field of Communication. Hardly surprising, much of the growth of DSP is being driven by communication. Thus, this gives DSPs a wide area to exploit.

In my final year, I joined hands with three other students to undertake a project. Our project consisted of designing an Automated Teller Machine (ATM). The heart of our system consisted of the 8086 microprocessors. We shall use assembly language for programming the microprocessor. This is one of the most interesting large projects that I had a chance to be a part of. It is an extremely practical experience, which I really enjoy.

During my undergraduate course in the Department of Electronics I have consistently scored above 65 %, which is a matter of great pride for me, as very few student have achieved this feat. This makes me several cuts above the commonplace. Our courses were purely theoretical in nature. I would like to change that. As I have mentioned below, I am a practical person and like to see things work.

I enjoy teaching and although I have no formal experience, my naturally strong communication skills will help me considerably. I envisage myself as a teacher and researcher – either in industry or in the university. I see myself in the role of a teacher to undergraduates. In fact, that's something I would enjoy immensely.

I am a very practical person, and my understanding is prodded by seeing things work. I am a quick thinker and learner. Your university could be the nucleus to accomplish my goals, as ample opportunities exist for students like me to initiate independent study and to become involved in active research programs, both experimental and theoretical. I would be more than happy to be given the opportunity to pursue my graduate study in your university. I am confident that in return for this I can make a perennial contribution to your institution.

STATEMENT OF PURPOSE

I would like to introduce myself as XXX. I have completed my Bachelor of Engineering in Electronics & Communication from XXXX VISVESVARAYA INSTITUTE OF TECHNOLOGY an esteemed institution under XX UNIVERSITY. I always had the ambition of pursuing the MS program right from my childhood days.

I firmly believe that inquisitive and explorative attitude leads to constant learning process. As an Electronics Engineer, I look for graduate study to refine my knowledge and skills in my areas of interest. I believe it will also serve to give direction to my goal of a career as research-oriented organization.

During my undergraduate study, subjects like Communication systems, Signal processing, VLSI, Control systems, Microprocessors have been my favorite subjects. In particular I like Signal processing & VLSI and would like to specialize in that. The electronic chip has always fascinated me. As years roll by the size of the chip is decreasing and the cost is decreasing effectively. Though the size is decreasing, the sophistication of the function performed by the chip is increasing all the time. This advancement in the technology of the chip industry has inspired me to work in that industry, but as for my MS degree, I would remain open to other topics as well.

My project work during my final year of undergraduate course was on Flash Microcontrollers (ATMEL AT89C55). The title of the project was "SERIAL COMMUNICATION BETWEEN CONTROL CARDS IN (10+1) MUX". 10 is the number of Voice/data channels. 1 is order wire which is not included in the voice/data channels and must be written separately. This may appear a deviation from the area in which I wish to specialize, I am aware of the role that Microcontroller plays in field of Electronics. Hence, with an aspiration to specialize in area of VLSI or Signal processing, I present my application in front of you hoping for positive consideration.

I am aware universities in USA offer great scope and encouragement for Intellectual & academic development. The process of technical advancement is very fast in USA. Hence, USA has emerged as world leader

in Science & Technology. To keep myself updated with latest technical advances I am keen to obtain MS degree in USA.

After consulting my Professors, Friends, seniors studying in USA your university has been recommended very highly for its excellent faculty & learning experience. I always believe that essence of university education lies in synergetic relationship between student and his department. I feel that graduate study at your university will be the most logical extensions of my objectives. I am confident that wholesome education that I receive at your university will stand me in good stead through out my career.

I sincerely request you to consider my application for admission into your reputed university. I would be grateful to you if I am accorded the opportunity to pursue my higher studies and given the opportunity, I am sure that I will be able to able to achieve my goals keeping in touch with the expectations of the university.

Looking forward to being a part of your graduate program.

DATE:

STATEMENT OF PURPOSE

Time and effort if put together, leads to success. I still remember that Friday night, I was debugging a hospital management project. I didn't realize how the night drifted off so quickly, but finally I arrived at the solution. These comprehensive efforts were aptly rewarded when my project was implemented for a cardiac clinic. This experience in developing a client server-based system infused into my mind the love for computers over electronics and aspired me to delve into the field of software engineering.

70% in my secondary school admitted me into Electronics & Telecommunication Engineering at one of the ranked colleges in Mumbai, Vivekanand Education Society Polytechnic (VESP). Here I realised the importance of Microprocessors, Satellite Networks, and Mobile Communication etc. in today's environment. The final year project "Cordless FM transmitter and Receiver" was concentrated on wireless communication.

Knowing the importance of computers in today's world, I enrolled into NIIT, a pioneer in providing software education in India. Here I learned operating systems like UNIX, Win NT, Linux, Programming languages like c, c++, Java, developer tools like Visual Basic, Microsoft Foundation Classes, DBMS like SQL server, Sybase and 32-bit windows architecture. The final semester project was to develop & to simulate day to day operations in a hospital. This project took me to the top of the class where the project was much appreciated by the centre head and the dean himself, because it facilitates all the functionalities that can be found in a medium sized hospital. My teachers encouraged me to present this project to the hospital manager of Varun Cardiac Clinic (VCC). Finally, after 2 months of interaction with doctors and managers of the hospital, I developed a real lifetime system by employing 3 assistant programmers. I got plenty of experience not only in dealing with clients, but also in work management and time management, as I was simultaneously studying in the diploma college too.

The natural progression after diploma was pursuing a Bachelor of Engineering degree in Electronics at Shah & Anchor Kutchi Engineering College. Electrical Networks, Electronics, Instrumentation and Control Systems, Robotics, Data and Digital Communications all gave me a sound idea of the technological advances in the modern world. My final year project "Inventory and Stores Control", developed in a GUI environment focussed on the inventory system. My joy knew no bounds when I heard that the project was successfully tested by experts and is replacing the older version in the college.

I believe that experience plays a key role in one's career. I was lucky enough to meet XX, a Senior ERP consultant at Oracle Apps, Japan, over Internet who gave me an opportunity to work on a live project for DSP Meryll Lynch Global Integrated Solutions Ltd as an offshore development from Feb 99 to Jul 99. Here I learned, how the subdivision system software of a huge company is integrated under a common head, so that the system works smoothly and leads to an efficient outcome. Working with international members of the team was simply fantastic. Although I programmed on a small part of a huge system, I played an important part as a team member who was always ready for more work. This has always been my motto "Work is Worship". These experiences had tuned me to the workings of a large corporate and have also polished my computer know-how. I am extremely proud to state that I am the only one from the entire college who has worked on an international level.

Masters from your University will not only impart a sound education in latest technologies but will also push me ahead in the computing world. My intentions, my dreams, my goals are clear, I look forward to becoming an ERP Consultant with software giants like Arthur & Anderson Consulting, Oracle Apps. etc. and I am sure a good education, infrastructure and opportunities that come my way, will help me in achieving my goals. I have always tried my best to achieve optimised and unconventional solutions to build a rising success.

Motivation Letter

I, XXX, am applying to University of XX for the admission to master's Program in Electrical Engineering with specialization in the field of Communications. Here is my brief statement of purpose.

Why Higher Education?

Right from my High school days I had a strong inclination to be an Electronics & Communication Engineer. My Interest in the subject has grown with each passing year and it has reached where even Ph.D. will not quench my thirst for knowledge. I have come to terms with the fact that there is an enormous proportion of subject to be assimilated, therefore I want to take up a research-oriented course for my career. These coupled with a background of my educated father-an Superintendent, mother –a lecturer and my brother, an Electronics & Communication Engineer has made my choice of higher studies all the more obvious.

Academic Background

I got admitted to Sir. XX Institute of Technology, one of the reputed institutions in the state through an entrance exam & by merit. I took Electronics & Communication course as my major for it offered me innumerable choices of specialization. During these 4 years I learnt Subjects like-Electronics Devices & Circuits, Pulse & Digital Circuits, Programming in C, Digital Integrated Circuits, Digital Communication Theorems Design, Computer Communication Networks and Digital Signal Processing. These subjects aroused my research instincts when I realized I am really excited about the prospects of me being a part of such research.

Research Background

I always had a liking for the books from a very young age, perhaps this made me inquisitive about everything I came across yearning to know it's intricacies. This later helped me immensely when I started representing my school team regularly at inter-school science festivals.

My first brush with serious research was in final year of my undergraduate course when I successfully completed a project titled **“Design and implementation of ARQ Protocols in Data Communication”** at Communication division, Electronics & Radar Development Establishment (L.R.D.E), Defense Research & Development Organisation, Government of India-Ministry of Defense, Bangalore as part of a 4-member team. The project implements ARQ Protocols in data Communication networks using the widely popular 80c31 microcontroller. The 80c31 systems used for Transmission and reception have been designed and implemented. We did

parallel research along with our guide providing valuable inputs apart from helping him with calculations.

Goals

In a decade from now I envisage myself as a Research Scientist in premier research Laboratory or as a professor in a top university. I understand that M.Sc. culminating in a Ph.D. is very essential, henceforth my short-term goal is to do M.Sc. in a good University.

Why University of XX?

A Graduate education that would enable me to pursue advanced degree is what I have aspired for. I also believe that good graduate program is an essential step for realizing my long-term goals. I hope to fulfill my ambition at University of XX, which I consider one of the best to pursue graduate studies. I have gone through the brochure and website very carefully and I have concluded that with excellent research facilities and highly knowledgeable faculty, University of XX will provide a perfect environment to focus all my resources towards research.

I am confident that I'll be able to meet the standards of the University. I am eagerly waiting for an opportunity to make significant contribution in research and Development programs of the University.

Conclusion

I fully understand the kind of dedication, perseverance and resolve a graduate program calls for; I believe that I have the ability and determination to tread on this exciting path.

In conclusion, I do hope that I'll be given a fair chance to realize my objectives by being granted an admission. I look forward for a long and mutually beneficial relationship with the University.